

opus2txt User Manual

Purpose

`opus2txt` converts selected data blocks from Bruker OPUS spectroscopy files into simple tab-separated text files suitable for plotting, fitting, spreadsheet, or scientific-analysis software.

This manual is available both as `MANUAL.md` and as `MANUAL.pdf`. The PDF copy is regenerated automatically whenever the Markdown manual is updated.

Download and start the application

Pre-built desktop applications are available from the latest GitHub Release. These packages include the required runtime and dependencies, so Python, PySide6, and `brukeropus` do not need to be installed separately.

Available packages:

- **Linux x86_64** — **AppImage**
- **Windows x86_64** — **standalone .exe**
- **macOS Apple Silicon** — **.dmg**
- **macOS Intel** — **.dmg**

Older versions remain available from the complete GitHub Releases archive.

Linux

Download the AppImage corresponding to the current release. If required by the desktop environment, mark the file as executable in its file properties, then open it normally.

The Linux AppImage is cryptographically attested through the project's GitHub Actions build process.

Windows

Download the `.exe` file and open it normally. No installation is required.

The Windows build is currently unsigned, so Windows may display a security warning when the application is opened for the first time.

macOS

Download the `.dmg` matching the Mac architecture:

- `arm64` for Apple Silicon Macs.
- `x86_64` for Intel Macs.

Open the DMG and launch **opus2txt**. The macOS builds are currently unsigned, so macOS may display a security warning when the application is opened for the first time.

Workflow

1. Select the output folder

Press **Select and convert**. The first dialog asks for the output directory. On first launch the dialog starts from the user's home directory. After successful conversions, **opus2txt** remembers the location of the most recently processed OPUS input file for future launches.

Paths containing hidden components (names beginning with `.`) are rejected.

2. Select OPUS files

After choosing the output directory, a second file dialog opens starting from that directory. Select one or more Bruker OPUS files with extensions `.0`, `.1`, `.2`, `.3`, or `.4`.

Hidden files and unsupported extensions are ignored.

3. Generated files

For each selected OPUS file, **opus2txt** creates up to three output types.

ABS If the OPUS file contains an absorbance block (**a**), data are written to:

ABS/<original_name>_ABS.txt

with columns:

wavenumber	absorbance
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SRay If the OPUS file contains a single-ray block (**sm**), data are written to:

SRay/<original_name>_SRay.txt

with columns:

wavenumber	transmittance
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METADATA Parameters printed by **brukeropus** are written to:

METADATA/<original_name>_META.txt

Existing output directories

The **ABS**, **SRay**, and **METADATA** directories are created automatically when needed. Existing directories are reused.

Errors

If an OPUS file cannot be read or converted, the application displays an error message identifying the file. Other selected files continue to be processed.

Running from source

Running from source is optional and intended for users who want to inspect, modify, or develop the program.

Dependencies

Python ≥ 3.10
PySide6
brukeropus

Virtual environment

Using a dedicated virtual environment:

```
python3 -m venv ~/.venv/opus2txt  
~/.venv/opus2txt/bin/python -m pip install PySide6 brukeropus  
~/.venv/opus2txt/bin/python opus2txt.py
```

pipx

With a recent version of pipx:

```
pipx run ./opus2txt.py
```

Platform support

The graphical interface uses PySide6/Qt. Qt requests the native platform file dialog when available, allowing integration with Linux desktop environments and the normal system dialogs on Windows and macOS.

Citation

For scientific use, please cite the archived Zenodo release corresponding to the version used. See the Citation section of the README and CITATION.cff.